Python Exception Handling

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Python :: Exception Handling...

Python provides two very important features to handle any unexpected error in your Python programs and to add debugging capabilities in them:

- **Exception Handling**: This would be covered in this tutorial.
- **Assertions**: This would be covered in Assertions in Python tutorial.

What is Exception?

- An exception is an event, which occurs during the execution of a program, that disrupts the normal flow of the program's instructions.
- In general, when a Python script encounters a situation that it can't cope with, it raises an exception. An exception is a Python object that represents an error.
- When a Python script raises an exception, it must either handle the exception immediately otherwise it would terminate and come out.

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Exception Type	Description	Examples of Exception Names
Syntax Errors	Python parser finds invalid syntax. Blocks code from running.	`SyntaxError`, `IndentationError`
Exceptions	Code has the right syntax but gets an error that breaks code execution during runtime.	`ZeroDivisionError`, `ValueError`, `TypeError`, `ModuleNotFoundError`
Logical Errors	No syntax or runtime error, but the logic is flawed, leading to incorrect results.	N/A (logical errors do not raise exceptions)

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Check standard Exceptions at
https://www.python.org/doc/essays/stdexceptions/

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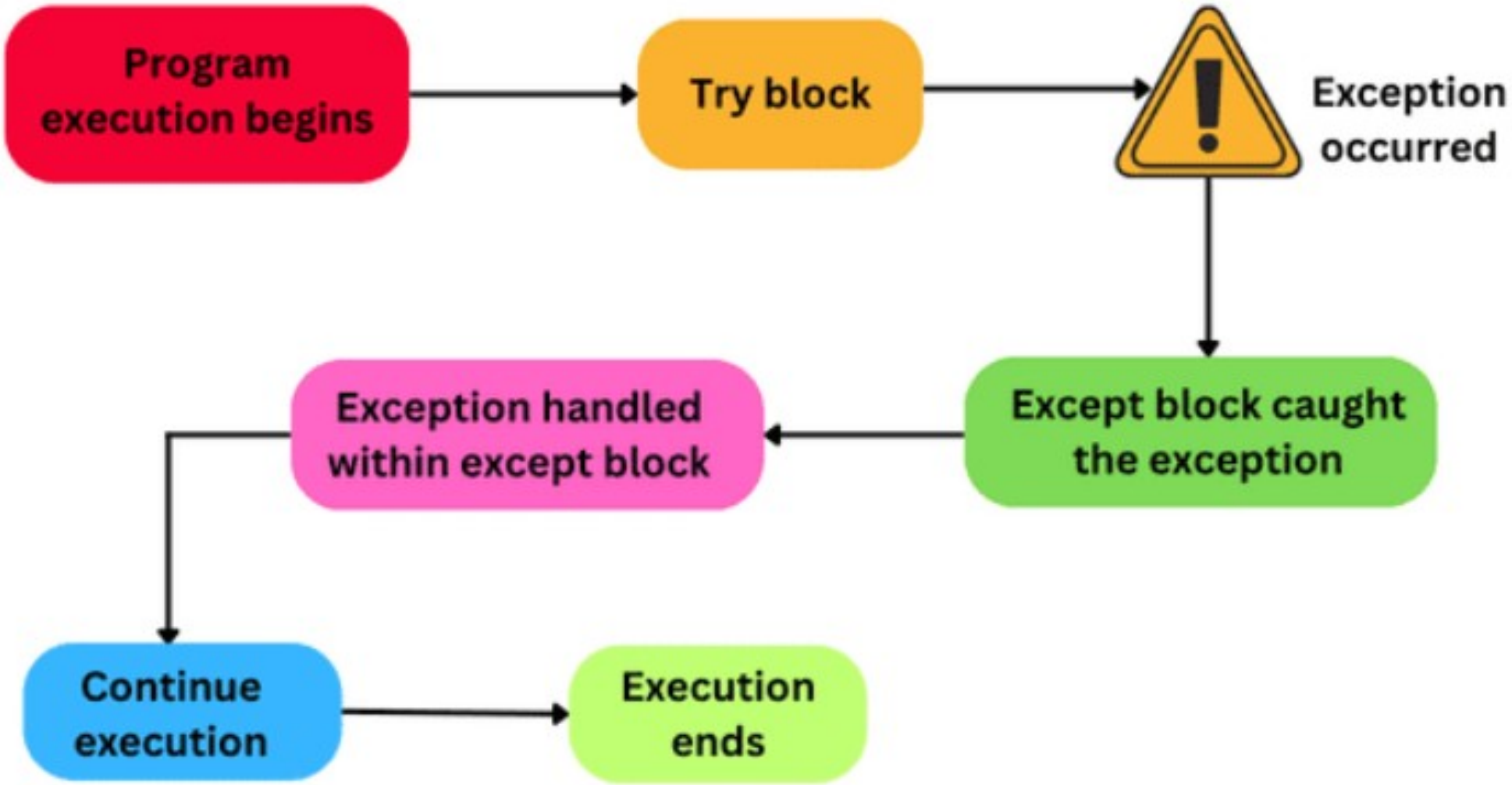

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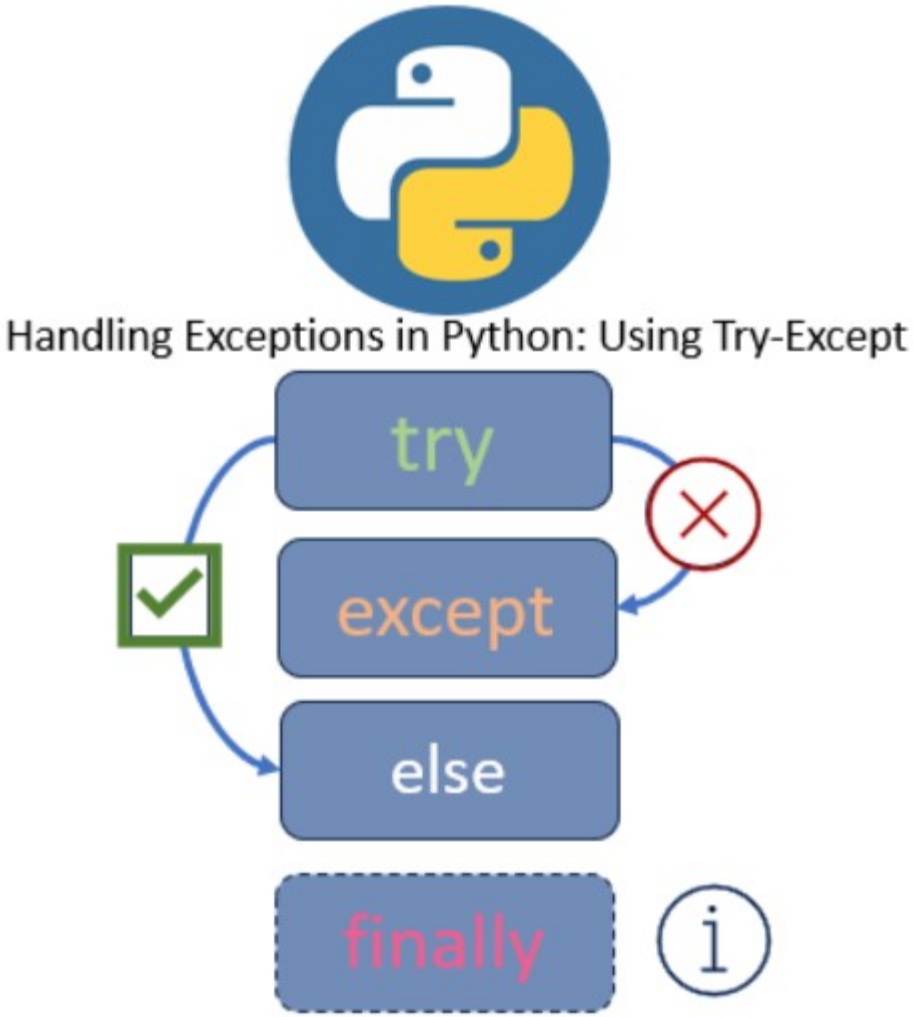

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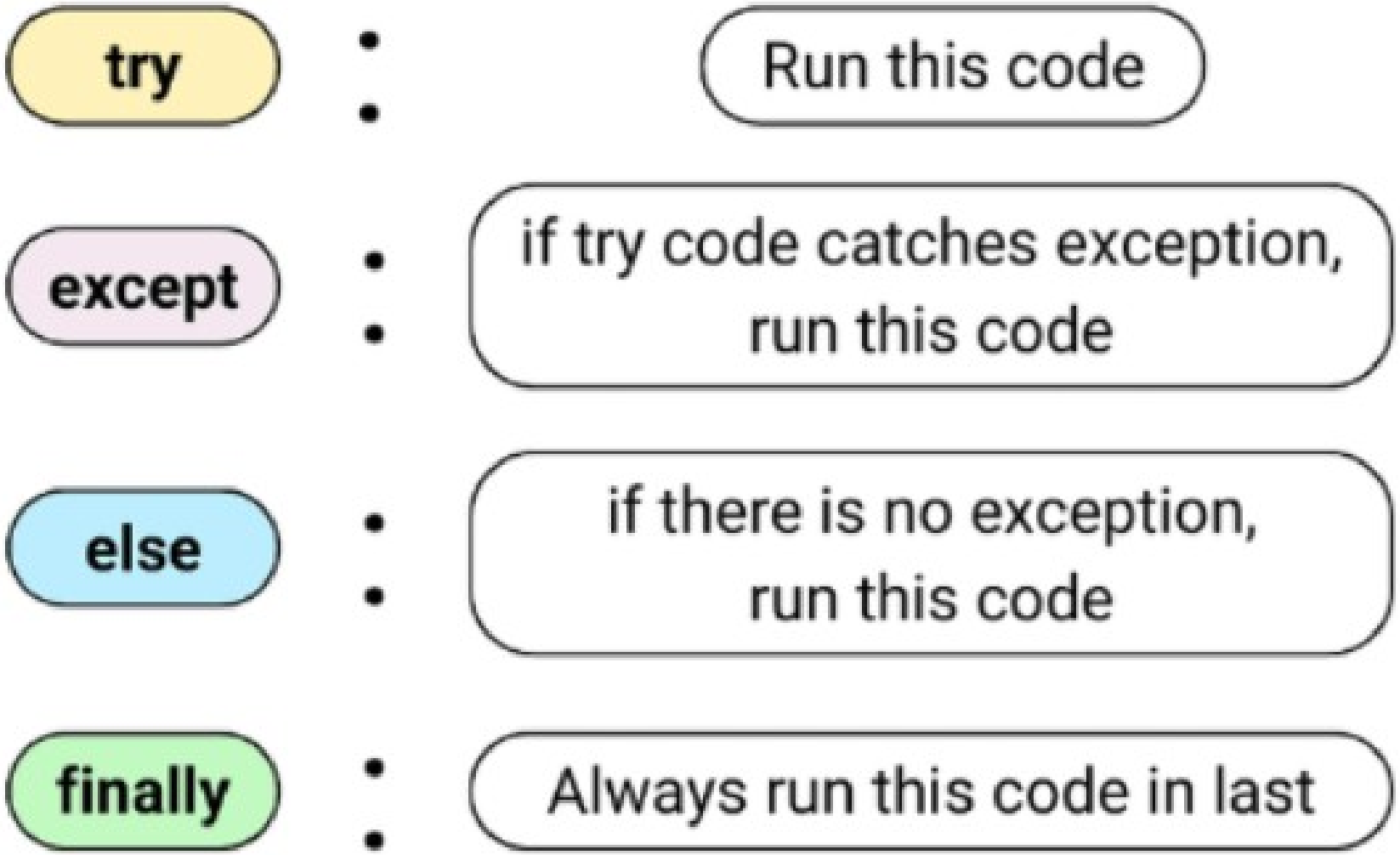

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```
try:
    statements
        ...
except:
    statements
        ...
else:
    statements
        ...
following_statement
```

Try

Run this as a normal part of the program

Except

Execute this when there is an exception

Else

Execute this only if no exceptions are raised

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```python
def divide(x, y):
    return x / y

def calculate(a, b):
    try:
        result = divide(a, b)
        print("Result:", result)
    except ZeroDivisionError:
        print("You can't divide a number by zero!")

calculate(10, 0)
```

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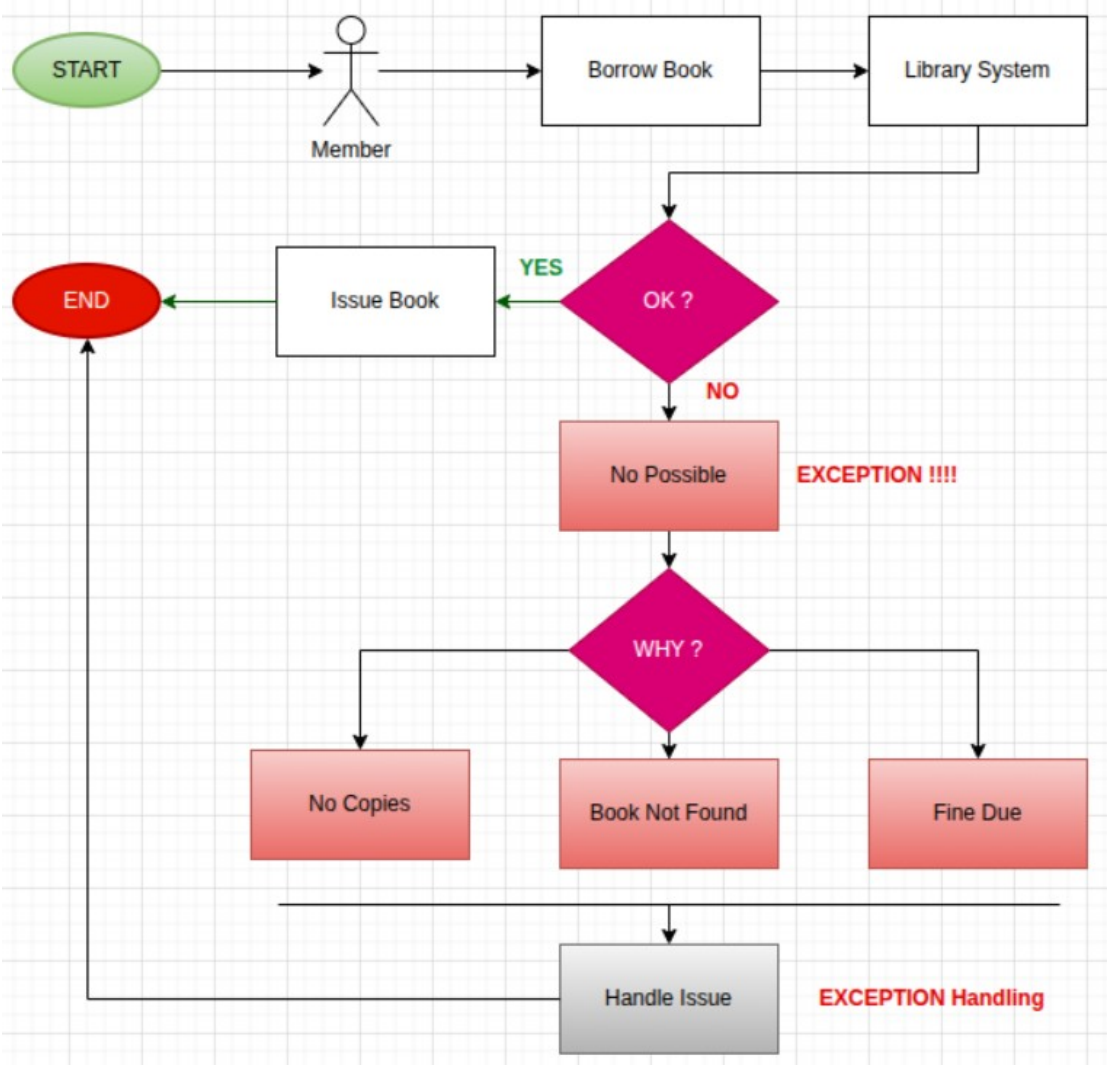

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```
try:
    book = library.get_book("Harry Potter")
    library.issue_book(student_id, book)

except BookNotAvailableError:
    print("😢 Sorry, no copies left!")

except InvalidStudentIDError:
    print("🪪 Hey! That ID looks suspicious!")

except FineDueError:
    print("💰 Pay your fine first!")

except Exception as e:
    print("🤯 Something weird happened:", e)
```

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